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Paleomagnetic study on the Triassic rocks from the Lhasa Terrane, Tibet, and its paleogeographic implications

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ABSTRACT

We present paleomagnetic results from the newly discovered Early-Middle and Late Triassic marine sediments of the Lhasa Terrane. Orientated samples were collected from 32 sites (330 samples) on the north side of the Dibu Co Lake (84.7°E, 30.9°N), Coqên County, in the western region of the Lhasa terrane. Rock magnetic data revealed that most of the samples were dominated by magnetite and/or pyrrhotite. The stepwise demagnetization curves illustrated three-components: a low temperature component (Component A) near the present-day field (PDF), a secondary remanent magnetization (Component B) that may be from the Cretaceous Period, and a high-temperature component (Component C). The Component C were isolated from the Early-Middle Triassic rocks in 8 sites (47 specimens) and from the Late Triassic rocks in 6 sites (37 specimens). The Component C of the Early-Middle Triassic rocks passed a reversal test (B class, 95% confidence level) and a fold test (99% confidence level), that of the Late Triassic rocks passed a fold test (95% confidence level). The corresponding paleopoles for the Early-Middle and Late Triassic periods of the Lhasa Terrane were at 18.9°N, 208.4°E with A_{95} = 3.9° and 19.6°N, 211.8°E with A_{95} = 10.7°, respectively. We suggest that the Lhasa Terrane maintained a relative stable latitude (16.5 ± 3.9°S and 18.4 ± 10.7°S) in the southern hemisphere during the Triassic Period before moving northwards and amalgamating with the main body of Eurasia. The Qiangtang and Lhasa terranes, which were located at the mid-low latitudes of the southern hemisphere, might have been isolated between Eurasia and Gondwanaland since the Early Triassic Period. The Meso-Tethys, potentially represented by the Bangong–Nujiang suture zone (BNS) between the Lhasa and Qiangtang terranes, opened up in the Early-Middle Triassic Period and expanded during the entire course of the Triassic Period.

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1. Introduction

The Tibetan Plateau is generally acknowledged to belong to the eastern Tethyan domain, its formation and evolution have long been a research focus for geologists (such as Chang and Zheng, 1973; Huang and Chen, 1987; Pan et al., 1997, 2012; Huang et al., 2000; Ren and Xiao, 2004; Li, 2008; Li et al., 2009; Wang et al., 2010; Zhu et al., 2011a,b; and Xu et al., 2011, 2013). It is generally accepted that the formation and evolution of the Tibetan Plateau were closely related to the rifted and northward-drift of a series of Gondwana-original terranes towards the southern margin of the Eurasia plate from the Late Paleozoic Period onwards, accompanied by the gradual contraction and/or opening of Paleo-, Meso-, and Neo-Tethys (Metcalfe, 1996, 2013; Wang et al., 2007; Ding et al., 2014).

The Lhasa Terrane is located between the Himalayan Terrane to the south and the Qiangtang Terrane to the north (Fig. 1). Recent investigations have suggested that the Qiangtang and Lhasa terranes preserve a large number of geologic records for the Tethys, the Bangong–Nujiang suture zone (BNS) between the two terranes most likely representing the disappeared relics of the Meso-Tethys (Fig. 1) (e.g. Xiao et al., 1986; Ren, 1997; Pan et al., 2004; Shi, 2007; Wang et al., 2008; Fan et al., in press; Liu et al., in press; and so on). This suture becomes a key region for addressing the process of amalgamation not only between the Qiangtang and Lhasa terranes but also between Gondwana and Eurasia. However, despite decades of mapping and data collection from this complex and long-lived convergence zone, the paleogrographic evolution of the Bangong–Nujiang ocean is still little understood, a factor which in turn places limitations on scientists' ability to produce an

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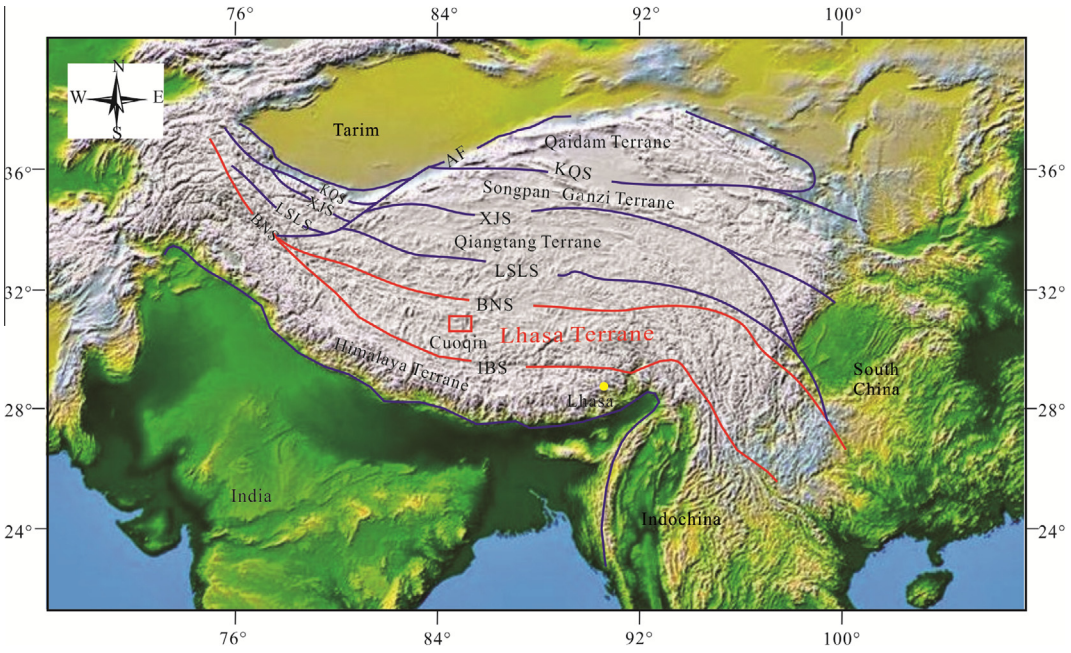


Fig. 1. Sketch of the major tectonic units of the Tibetan Plateau. IBS: Indus-YarlungZangbo suture zone, BNS: Bangong-Nujiang suture zone, LSLS: Lungmu Co-Shuanghu suture zone, XJS: Xijir Ulan-Jinshajiang suture zone, KQS: Kunlun-Qinlin suture zone, AF: Altyn Tagh Fault. Red box: study area. (modified after the Geological Map of Qinghai-Xizang (Tibet) Plateau and Adjacent Areas, Chengdu Cartographic Publishing House). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

accurate paleogeographic reconstruction of the Tibetan Plateau (Besse and Courtillot, 2003; Stampfli and Borel, 2002; Collins, 2003; Scotese, 2004; Golonka, 2007; Torsvik et al., 2008; Metcalfe, 2009; Choulet et al., 2011).

Paleomagnetism is an effective approach to the past tracks of plates and terranes. Paleomagnetic results can determine the paleolatitude of a terrane and, relevant changes thus can be applied to the paleogeographic reconstruction of the plate. In recent years, a series of paleomagnetic studies have been conducted on the Lhasa Terrane of the Tibetan Plateau and some important achievements have been accomplished (e.g. Pozzi et al., 1982; Achache et al., 1984; Zhu et al., 1984; Ye and Li, 1987; Lin and Watts, 1988; Dong et al., 1991; Torsvik et al., 2009; Tan et al., 2010; Yi et al., 2011; Sun et al., 2012; Liebke et al., 2013; Chen et al., 2014; Yang et al., in press; Zhao et al., 2015; and so on). Owing to different concerns (these studies mainly focus on the India–Asia collision), the available paleomagnetic data is mainly distributed in the Tertiary and the Cretaceous periods of Lhasa Terrane. There is little available data from the Paleozoic Period and none from the Early Mesozoic Period (Dong et al., 1991; Guo, 2009; Cheng, 2012; Ran et al., 2012a). In short, the present published paleomagnetic data are insufficient for us to clearly determine the movement of the Lhasa Terrane before its amalgamation with the main body of Eurasia, and insufficient for us to constrain the successive closure and opening of the Tethyan basins in Tibetan Plateau.

In this paper, we describe two Triassic paleomagnetic results from the newly discovered Triassic limestone on the Coqên area of the Lhasa Terrane to address the limitations in previous data sets. By investigating the positional changes (including the absolute position and the relative positions between terranes) and the kinematic characteristics of the Lhasa and Qiangtang terranes during the Triassic Period, this study provides critical paleomagnetic evidence for studying the Tethyan evolution and the reconstruction of the Tibetan Plateau.

2. Geology setting and sampling

Previously, it was believed that Triassic strata were missing in large areas of the western Lhasa region or no one could tell with certainty whether Triassic strata had developed in the Lhasa Terrane (Li et al., 1982; Rao et al., 1987; Liu et al., 1990; Xia and Liu, 1997; Zhao et al., 2001; Zhou et al., 2002). Since 2004, Ji et al. have successively discovered Early Triassic conodonts (*Pachycladina*, *Neohindeodella, Cornudina, Hadrodontina* and *Hibbardella* sp., etc.) in the Shiquanhe area of the Lhasa Terrane (Ji et al., 2006, 2007a; Zheng et al., 2007). Later, not only the Late Triassic Norian *Epigondolella* (Ji et al., 2006) but also Early and Middle Triassic conodonts (*Paragondolella polyg-Nathiformis*, *Xaniognathus* cf. *newpassensis* and *Iranognathus* sp., etc.) were observed (Wu et al., 2007) on the northern margin of the Dibu Co Lake in the Coqên area of Lhasa Terrane (Fig. 2a and b). The biostratigraphic studies of conodonts have successfully distinguished the Early to Late Triassic marine carbonate formations from the strata which was initially acknowledged to be the Permian in age. A lithostratigraphic unit division of Triassic strata in the Coqên area has been preliminarily proposed. According to the lithological features, the strata are divided (from bottom to top) into the Dibu Co, Gyangrang, Zhuglung, and Garing Co Formations (Table 1) (Ji et al., 2007b). Each formation is mainly composed of marine carbonates, and the content of these siliceous deposits is the primary marker for distinguishing between various formations. Siliceous deposits are developed locally in certain horizons of the Lower Garing Co Formation, which are mostly well developed in the Middle Zhuglung Formation and undeveloped in the Upper Gyangrang Formation. The Garing Co Formation is most likely in conformity with the Late Permian strata. The top boundary of the Gyangrang Formation contains dolomitic limestone and is in unconformity with the clastic rocks of the overlying Dibu Co Formation (Fig. 2d), which palynological studies have confirmed is composed of Late Triassic – Early and Middle Jurassic strata (Zhou et al., 2002; Ji et al., 2007b).

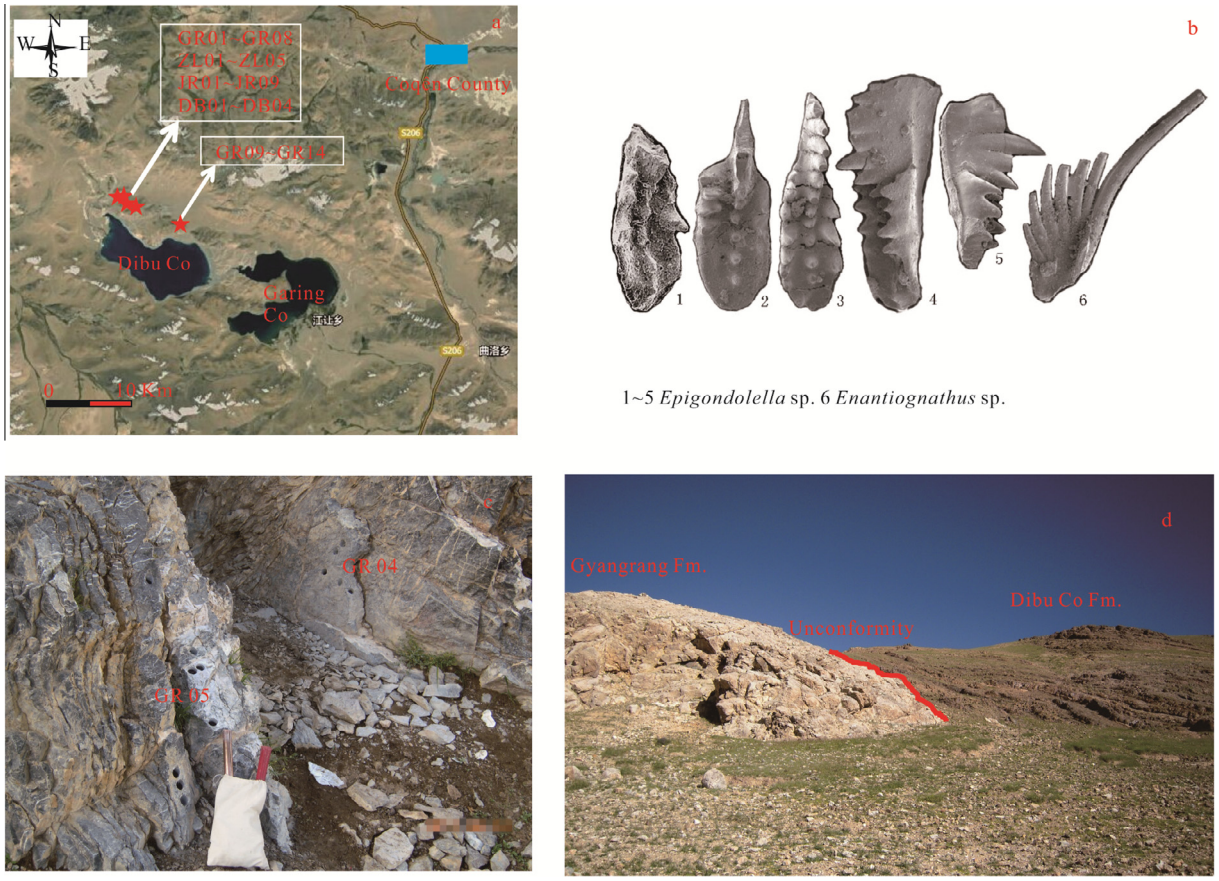


Fig. 2. GPS map of the sampling locality (a), conodont fossils in the Late Triassic Gyangyang Formation of the section (Published from the Ji et al., 2006) (b), photographs showing the nearly-upright strata in Garing Co formation (c), Unconformable contact between the Gyangrang Formation and the Dibu Co Fomation (d). Red five-pointed star: sampling section.

Table 1
Stratigraphic sequence of the sampling region (according to Ji et al., 2007a,b).

Geologic time		Sampling strata of the Lhasa Terrane	
Triassic	Late Triassic	Rhaetian	Dibu Co Formation [a] (T_3db)
		Norian	GyanGRang Formation [a] (T_3jr)
		Carnian	Zhuglung Formation [a] (T_3zl)
	Middle Triassic	Ladinian Anisian	Garing Co Formation[a] ($T_{1-2}gr$)
	Early Triassic	Olenekian Induan	
Permian	Late Permian	Changhsingian	Angjie Formation

[a] Sampling formation.

With the aid of a portable petrol-powered drill, individually oriented paleomagnetic samples were collected in 32 sites (330 samples) from the Triassic formations on the northern margin (84.7°E, 30.9°N) of Dibu Co Lake in Gyangrang Town, Coqên County, of the central Lhasa Terrane (Figs. 1 and 2). The sampling included 4 sites (40 samples) of clastic rocks from the Dibu Co Formation (T_3db), 9 sites (77 samples) of limestone from the Gyangrang Formation (T_3jr), 5 sites (53 samples) of limestone from the Zhuglung Formation (T_3zl), and 14 sites (160 samples) of limestone from the Garing Co Formation (T_3gr). All paleomagnetic samples were oriented by magnetic compass.

3. Laboratory methods

To identify the major magnetic minerals in the samples for choosing appropriate demagnetization steps, a few representative specimens were chosen from each sampling site for systematic thermal demagnetization analysis of saturation isothermal remanent magnetization (SIRM) and three-axis IRM (Lowrie, 1990). A direct current magnetic field was applied to acquire the IRM with a 2 G Enterprises Pulse Magnetizer with a maximum field of up to 2700 mT. Using a JR-5A spinner magnetometer IRM acquisition was performed, while the systematic thermal demagnetization for the three-axis IRM was conducted, in a magnetically shielded room, using an MMTD-48 thermal demagnetization oven and a 2G-755R superconducting magnetometer.

According to the results of the rock magnetic analysis, we chose progressive thermal demagnetization, alternating field demagnetization or composite demagnetization (i.e., progressive thermal demagnetization and alternating field demagnetization with the maximum temperature at 300 °C) for the systematic magnetic cleaning of all specimens to obtain stable remanent magnetization components. Thermal demagnetization was conducted in an MMTD-48 thermal demagnetization oven at temperature intervals of 30–100 °C for the low-temperature (below 300 °C) and 10–25 °C for the high-temperature (over 300 °C). Progressive alternating field demagnetization was conducted with a 2G-755 alternating field demagnetizer at a demagnetization interval of 5–10 mT. The remanent magnetization measurement was completed on a 2G-755R cryogenic superconducting magnetometer. Specimen demagnetization and measurement were performed in a magnetic shield space with residual fields less than 300 nT in the central area. All of the experimental analyses were performed in the Paleomagnetic Laboratory at the Institute of Geology and Geophysics, CAS, Peking, China (Zhu et al., 2003).

Remanent magnetization components were isolated using Kirschvink's principle component analysis (PCA) (Kirschvink, 1980), while a few of specimens were processed using remagnetization great circles (GC) (Halls, 1978). The site-mean (specimens-mean) directions were calculated using Fisher's statistical mean (Fisher, 1953) of spherical unit vectors or the mixed mean of unit vectors and the remagnetization great circles (McFadden and McElhinny, 1988). The software PMGSC of R. Enkin (Geological Survey of Canada) was used for paleomagnetic data analysis.

4. Rock magnetic results

The rock magnetic results are shown in the IRM acquisition curves and the demagnetization curves of the IRM (Fig. 3). Initially, the IRM curves of the limestone specimens climb steeply and reach quasi-saturation at less than 300 mT, which suggests a fraction of low and medium coercivity magnetic minerals. The three intensity-decay curves of thermal demagnetization clearly exhibit drastic decays at about 320 °C, indicating the presence of moderate-unblocking-temperature magnetic minerals (e.g., pyrrhotite) in the samples. The soft and medium coercivity fractions finally decay to zero at approximately 570 °C, reflecting the characteristics of magnetite. Thus, we can conclude that in the marine carbonate formations of the Coqên area, the predominant magnetic carriers are pyrrhotite and magnetite.

5. Paleomagnetic results

A total of 335 specimens of Triassic rock from the Lhasa Terrane were subjected to systematic demagnetization treatment. For the

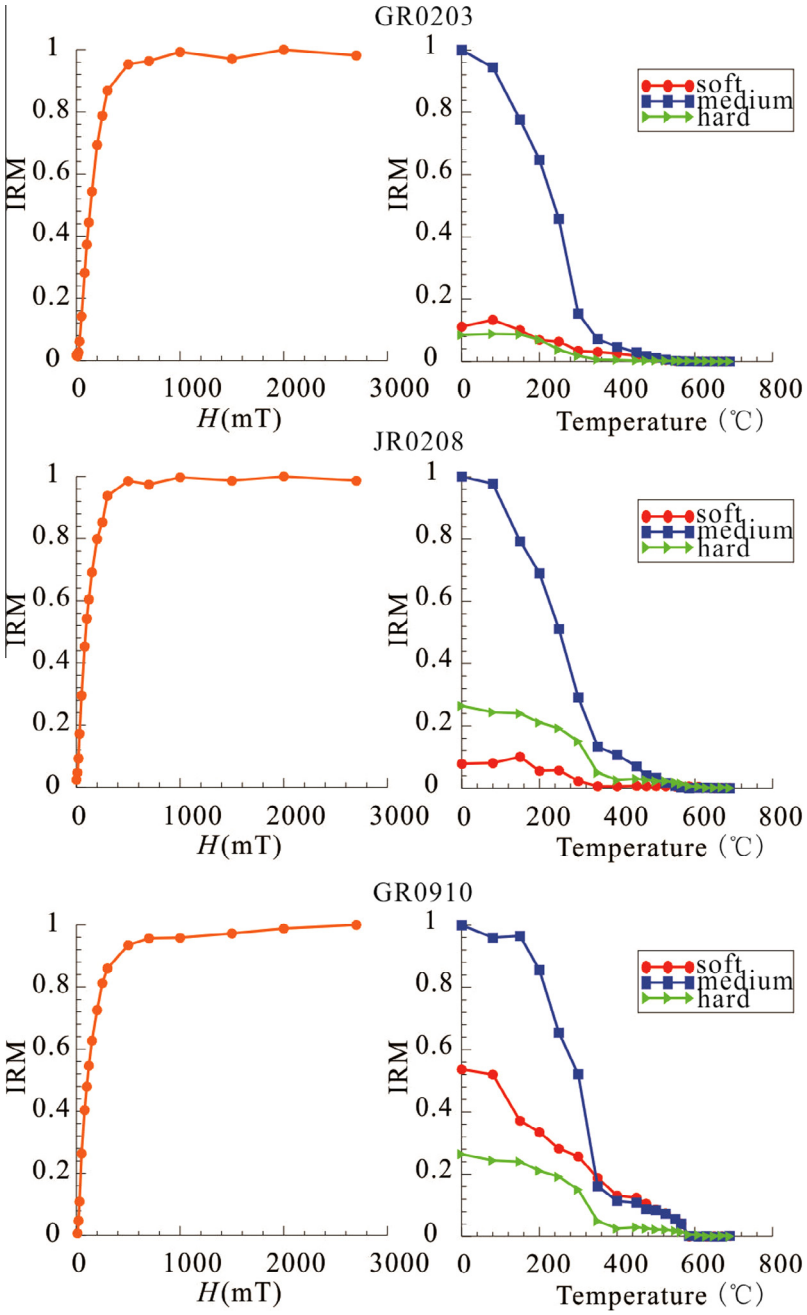


Fig. 3. Rock magnetic behavior of representative specimens: acquisition curves of IRM (left), and curves of thermal demagnetization of three-axis IRM (right) (Lowrie,1990). The soft (<40 mT), medium (40–400 mT), and hard (400–2700 mT).

majority of specimens, the intensity decay curves of the remanent magnetization showed significant decay at approximately 250–350 °C and underwent a change of 2–10% of the NRM at 400 °C. Combined with the results from the rock magnetic analyses, we could infer that magnetic minerals with a medium-to-low coercivity and a moderate blocking temperature, such as melnikovite or pyrrhotite, were unblocked. In these specimens, there is a viscous remanent magnetization (Component A) of the modern geomagnetic field in the initial stage of demagnetization (NRM-250 °C). Additionally, a stable remanent magnetization (Component B) was obtained at above 350 °C through principal component analysis, but this disappeared in the temperature interval of 400–550 °C (Fig. 4a and b).

For some of specimens from the Garing Co, the Zhuglung, and the Gyangrang Formations, the intensity decay curves of the remanent magnetization exhibit a gradual and slowly decreasing trend at temperatures below 500 °C. The intensity of the remanent magnetization suddenly rebounds and increases to a high level above 500 °C, and is completely unblocked between 565 and 585 °C. These characteristics suggest that the main magnetic carrier is magnetite. After Component A and/or Component B were isolated at temperatures below 500 °C, another remanent magnetization

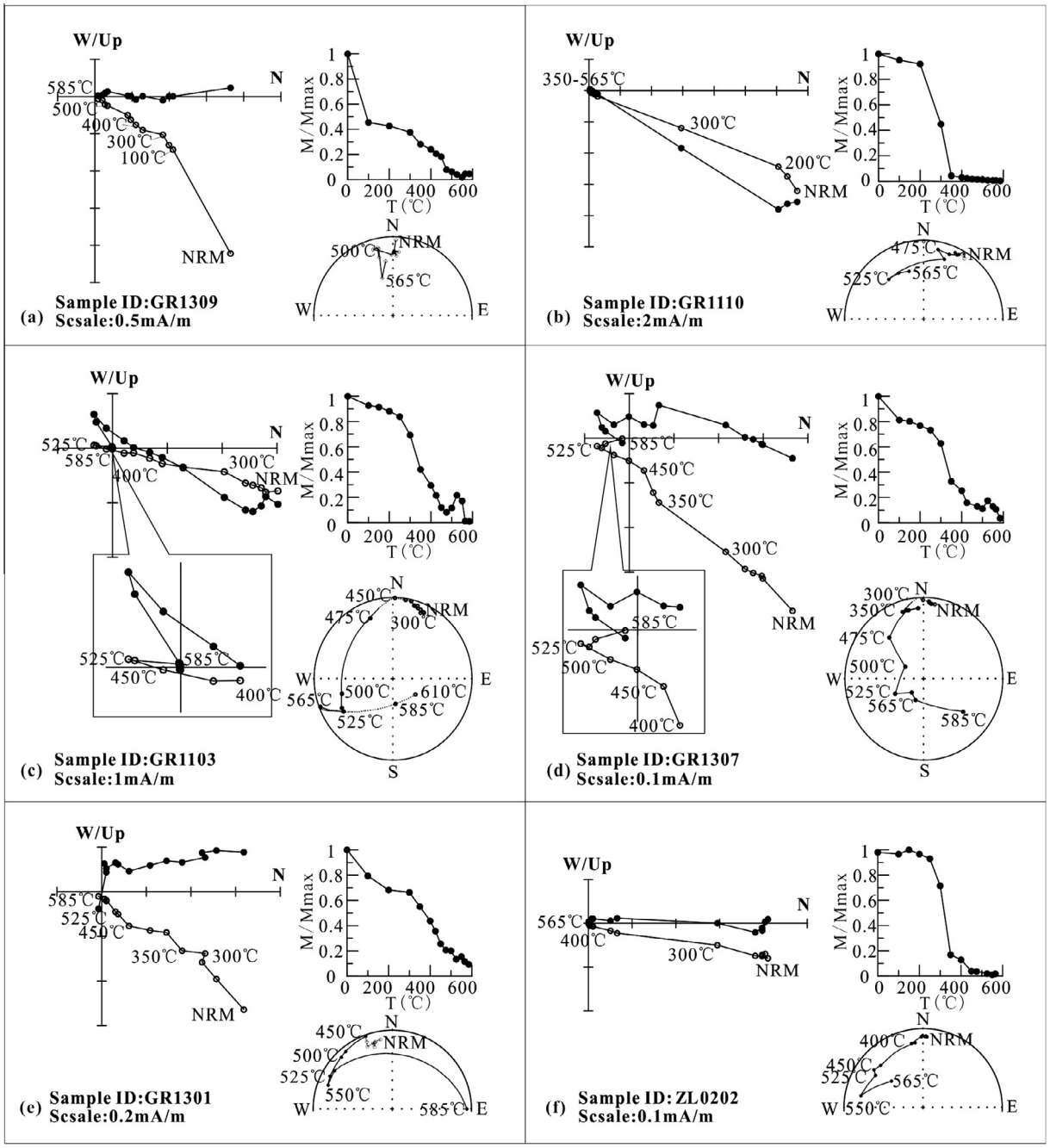


Fig. 4. Demagnetization diagrams showing results of thermal demagnetization in this study. Zijderveld diagrams (Zijderveld, 1967) are plotted in geographic coordinates, open circle: vertical plane projection, solid circle: horizontal plane projection. Stereographic projections are plotted in stratigraphic coordinates, Open circle: upper hemisphere projection, solid circle: lower hemisphere projection. Component A and Component B were isolated from representative specimens (a, b) representative specimens were isolated using Kirschvink's principle component analysis (c, d); representative specimens were isolated using remagnetization great circles (e, f).

(Component C) was obtained from these specimens at 525–585 °C (Fig. 4c and d). For some other specimens, the Component C was not easily isolated using Kirschvink's principle component analysis (Kirschvink, 1980), but a remagnetization great circle with sector constraints (Halls, 1978) was used to isolate the Component C (Fig. 4e).

5.1. Component A

The Component A (NRM-250 °C) was isolated from most of the specimens. The site-mean direction is $D_g = 354.9°$, $I_g = 42.5°$ ($k_g = 42.8$, $\alpha_{95g} = 4.5°$, $n = 25$ sites) in geographical coordinates (please see Table 1 in the Supplementary material for detailed

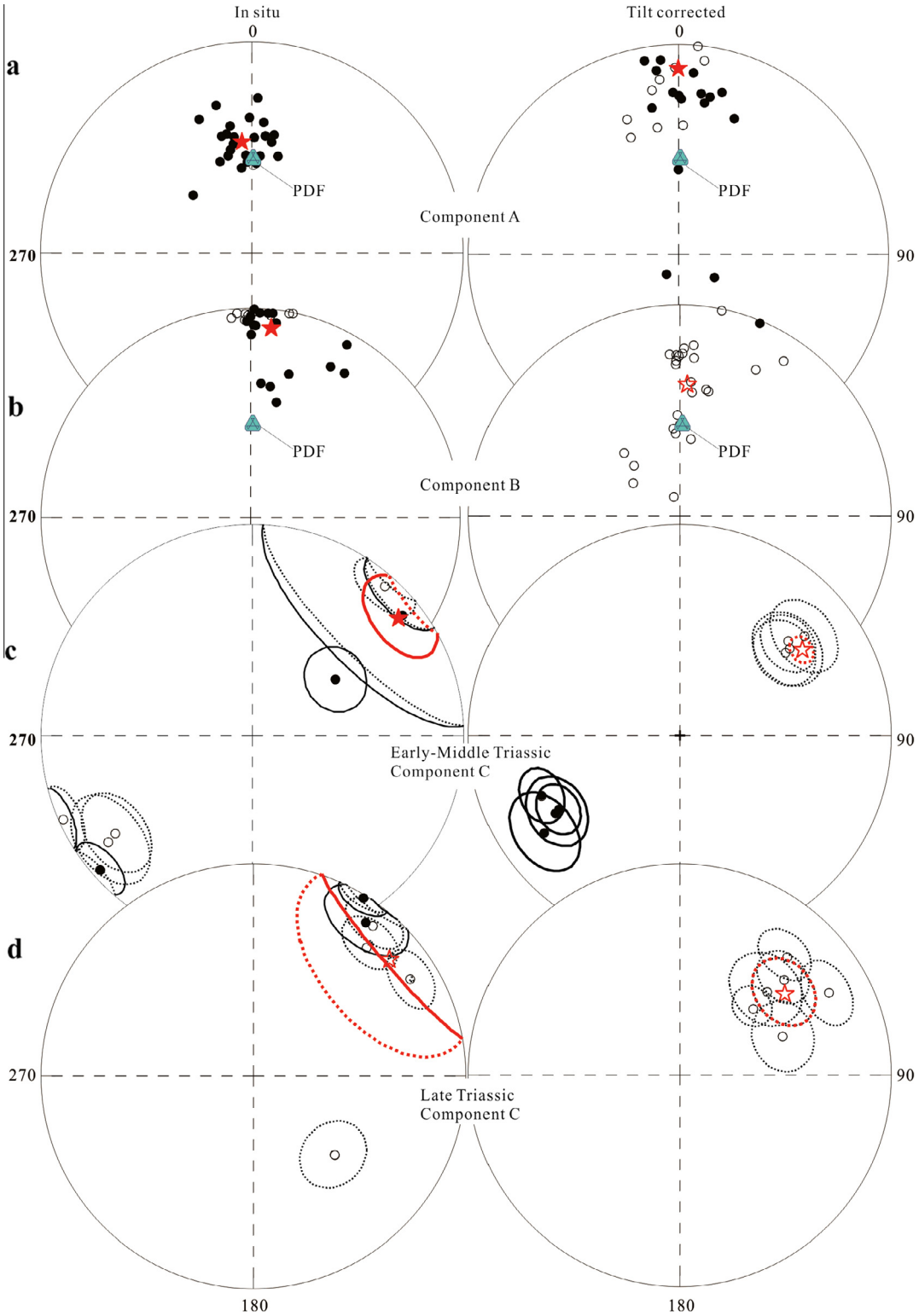


Fig. 5. Equal-area projections of site-mean directions of the Component A, the Component B, the Component C from Early-Middle Triassic rocks, and the Component C from Late Triassic rocks (a, b, c, d). Open circle: upper hemisphere projection, solid circle: lower hemisphere projection, Blue triangle (PDF): present-day Earth magnetic dipole field, Red five-pointed star: overall site-mean directions with α_{95} confidence limits (circles). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

information). The direction with a negative fold test (McElhinny, 1964; Watson and Enkin, 1993) is close to the dipole sub-field direction of the present-day field (PDF, $D = 0.4°$, $I = 50.1°$) (Fig. 5a). Thus, we propose that it possibly represents the viscous remanent magnetization of the present geomagnetic field.

5.2. Component B

After Component A was removed by magnetic cleaning at approximately 250 °C, a stable Component B was isolated between the temperature intervals from 250 to 500 °C. The Component B were isolated from 27 sites (237 specimens) (Table 2), the site-mean direction is $D_g = 6.2°$, $I_g = 8.1°$ ($k_g = 26.1$, $\alpha_{95g} = 5.5°$, $n = 27$ sites) and $D_s = 3.2°$, $I_s = -33.9°$ ($k_s = 10.6$, $\alpha_{95s} = 9.0$, $n = 27$ sites) before and after tilt corrections. The Component B fails the McElhinny's (1964) fold test and the McFadden's (1990) fold test, the Watson and Enkin's (1993) fold test gives an optimal concentration at 3.7 ± 5.5% unfolding percentage (Fig. 6), indicating a post-folding age of the magnetization. This component B is significantly different from the direction of PDF both in geographic and stratigraphic coordinates (Fig. 5b). The site-mean direction of the component B prior to tilt corrections corresponds to a paleomagnetic pole position at 62.5°N, 251.2°E ($dp/dm = 5.5/2.8$). This remagnetization pole is close to the Early Cretaceous paleopole of the Lhasa Terrane (66.4°N, 220.3°E ($dp/dm = 9.3/6.9$)) reported by Sun et al. (2008) and the Late Cretaceous paleopole (63.1°N, 224.6°E ($A_{95} = 5.1$)) reported by Tang et al. (2013). The Component B likely represents a secondary remnant magnetization formed in the Cretaceous Period, and its formation may be related to the Late Jurassic-Early Cretaceous orogenic movement (Kapp et al., 2007; Li et al., 2008) of this area.

5.3. Component C

We isolated the Component C in 8 sites (47 specimens) from the Early-Middle Triassic Garing Co Formation. The Component C exhibits both a positive inclination to the SW and a negative inclination to the NE in stratigraphic coordinates. Four sites yield a reversed-polarity high-temperature component ($D_s = 239.3°$, $I_s = 28.8°$ ($k_s = 147.8$, $\alpha_{95s} = 7.6°$, $n = 4$ sites), and the other four sites record a normal polarity $D_s = 51.0°$, $I_s = -32.2°$ ($k_s = 252.4$, $\alpha_{95s} = 5.8°$, $n = 4$ sites). We converted the reversed-polarity Component C into the normal polarity, and then used the standard Fisher-statistics to calculate the mean direction. The over-all mean direction of the eight sites is: $D_g = 51.2°$, $I_g = 10.2°$ ($k_g = 16.4$, $\alpha_{95g} = 14.1°$, $n = 8$ sites), and $D_s = 55.2°$, $I_s = -30.6°$ ($k_s = 137.1$, $\alpha_{95s} = 4.7$, $n = 8$ sites) before and after tilt corrections (Fig. 6c, Table 3; please see Table 2 in the Supplementary material for detailed information). This site-mean direction passed the reversal test (B-level reverse; $\gamma_0 = 7.9 < \gamma_c = 8.3$, at 95% confidence level) (McFadden and McElhinny, 1990), McElhinny fold test at the 99% confidence level (($k_s/k_g = 137.1/16.4 = 8.36$) > (F (14, 14) = 6.13)) (McElhinny, 1964),

Table 2
Paleomagnetic data for the Component B.

Site ID	n/N	In situ		Tilt corrected		α_{95}	k	Pole position	
		D_g (°)	I_g (°)	D_s (°)	I_s (°)			λp (°N)	ϕp (°E)
GR/01	12/15	10.7	−0.7	11.7	−35.0	3.6	146.3	57.2	244.7
GR/02	13/14	5.7	−2.6	5.6	−37.1	4.2	98.3	57.3	254.1
GR/03	12/13	4.9	1.8	4.7	−32.7	8.7	25.8	59.6	255.0
GR/04	6/8	7.5	5.6	317.6	−61.0	6.4	110.5	61.0	249.1
GR/5,7	7/19	11.7	−0.4	305	−65.7	8.9	46.9	57.0	242.9
GR/06	5/11	0.9	7.2	318.4	−54.2	8.4	83.9	62.7	262.7
GR/08	7/11	29.1	5.0	342.1	−81.5	11.2	30.0	50.5	214.9
GR/09	8/9	14.9	26.2	12.9	−35.6	4.5	152.4	68.1	222.7
GR/10	9/10	27.9	16.6	27.4	−19.8	8.0	42.4	55.7	209.6
GR/11	9/10	33	16.0	33.8	−10.6	4.3	144.3	51.7	204.2
GR/12	12/12	8.5	33.0	22.5	1.1	1.8	581.8	75.0	231.9
GR/13	10/12	4.4	32.1	5.1	−22.4	10.8	21.0	75.9	247.2
GR/14	16/17	12.6	39.2	11.5	−0.7	7.3	26.5	75.8	209.4
Site mean	13	13.2	13.9	6.7	−37.2	9.2 (17.3)	21.3 (6.7)	63.2	234.6
Zl/1,2	12/20	358.3	−5.0	358.6	−23.9	7.5	34.4	56.6	267.8
Zl/03	12/12	0.5	−1.8	0.9	−20.7	6.1	51.6	58.2	263.8
Zl/04	10/11	356.2	−2.0	356.3	−21.0	3.0	260.1	57.9	271.9
Zl/05	8/10	359.3	−3.0	359.6	−21.9	6.2	80.8	57.6	266.0
Site mean	4	358.6	−3.0	358.9	−21.9	2.6 (2.6)	1248.8 (1248.8)	57.6	267.3
JR/01	6/9	358.9	5.5	355.7	−52.7	8.8	58.9	61.9	267.1
JR/02	8/9	360.0	3.5	357	−54.8	9.0	38.8	60.9	264.7
JR/03	7/8	6.2	1.6	8.1	−56.9	8.2	55.1	59.3	252.5
JR/04	8/8	1.5	7.4	0.0	−51.1	6.0	86.2	62.8	261.4
JR/05	3/5	0.2	11.3	358.6	−47.0	12.8	93.8	64.8	264.2
JR/06	9/11	2.6	1.9	4.6	−17.0	6.5	63.7	59.9	259.5
JR/07	5/9	354.5	−3.8	358.4	−25.4	13.0	35.6	56.8	274.8
JR/08	7/11	1.0	0.4	1.4	−18.5	9.9	38.1	59.3	262.7
JR/09	7/11	358.5	−2.4	358.7	−21.4	7.1	73.2	57.9	267.5
Site mean	9	0.4	2.8	0.3	−38.4	3.6 (11.3)	205.4 (21.7)	60.5	263.9
DB/1,2,3,4	9/40	3.4	25.2	6.8	−3.2	9.3 (9.1)	31.6 (33.0)	72.1	253.9
All sites mean	27	6.2	8.1	3.2	−33.9	5.5 (9.0)	26.1 (10.6)	62.5	251.2

Notes: Site ID: site identification; n/N: number of samples used for the calculation/demagnetized or number of sites; D_g, I_g, D_s, I_s: remanence directions in geographic and stratigraphic coordinates; k : precise parameters; α_{95}: confidence angle. λp: latitude of the corresponding virtual geomagnetic pole (VGP) in stratigraphic coordinates; ϕp: longitude of the corresponding virtual geomagnetic pole (VGP) in stratigraphic coordinates.
(1) Watson and Enkin's fold test (1993): optimal concentration at 3.7 ± 5.5% unfolding percentage.
(2) McElhiny's fold test (1964): $N = 29$, $k_s/k_g = F\ (2 * (n\ 2–1), (n\ 1–1)) = 0.5$, with negative fold test.
(3) McFadden's fold test (1990): $\xi_2 = 6.677$ before and $\xi_2 = 17.322$ after tilt corrections, critical ξ, at 99% = 6.044 and 95% = 8.546 with negative fold test.
(4) Paleomagnetic pole: $\lambda = 62.5°$N, $\Phi = 251.2°$E, $dp/dm = 5.5/2.8$ (VGP calculated from the directions in the geographic coordinate).

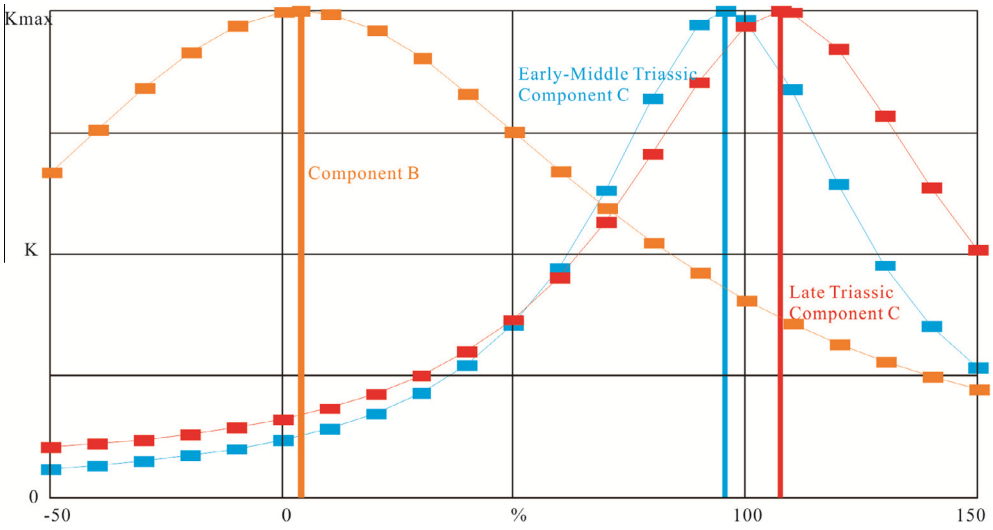


Fig. 6. Results of the Progressive correction test of the Component B (yellow line), the Component C of the Early-Middle Triassic (blue line), and the Component C of the Late Triassic (red line). *K*, corresponding the precision parameter. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
The Early-Middle Triassic Paleomagnetic data of the Component C.

Site ID	n_1,n_2/N	In situ		Tilt corrected		α_{95}	k	Pole position	
		D_g (°)	I_g (°)	D_s (°)	I_s (°)			λp (°N)	ϕp (°E)
GR/01	2,3/15	41.7	−5.8	49.7	−32.8	10.7	52.1	21.8	213.3
GR/02	1,5/14	46.2	0.7	51.4	−25.0	13.2	26.7	23.9	208.4
GR/3,5	2,3/19	45.2	2.4	51.3	−34.4	13.7	32.1	20.0	212.9
GR/6,7,8	2,4/35	56.1	50.6	51.7	−36.6	12.2	31.1	18.6	213.8
GR/10*	0,3/10	237.2	−8.1	232.7	31.7	7.2	49.9	20.1	210.7
GR/11	3,4/10	246.1	−2.0	246.3	29.0	9.5	41.3	11.3	200.6
GR/12	2,3/12	233.6	−16.6	238.2	31.6	12.5	38.4	16.3	206.9
GR/13	5,2/12	234.3	−21.1	234.4	21.6	13.9	19.8	23.0	204.7
GR/14	4,2/17	228.7	4.0	238.5	32.7	9.0	56.4	15.7	207.2
Normal	4	46.4	11.3	51.0	−32.2	31.2 (5.8)	9.6 (252.4)	21.2	212.1
Reverse	4	235.7	−9.0	239.3	28.8	16.0 (7.6)	33.9 (147.8)	16.6	204.8
All sites mean	8	51.2	10.2	55.2	−30.6	14.1 (4.7)	16.4 (137.1)	18.9	208.4

Notes: Site ID: site identification; D_g, I_g, D_s, I_s: remanence directions in geographic and stratigraphic coordinates; K : precise parameters; α_{95} : confidence angle. λp: latitude of the corresponding virtual geomagnetic pole (VGP) in stratigraphic coordinates; ϕp : longitude of the corresponding virtual geomagnetic pole (VGP) in stratigraphic coordinates; n_1,n_2/N: number of samples used remagnetization great circles, Kirschvink's principle component analysis for the calculation/number of sites; GR/10* discarded from the all-sites-mean average due to large 95% confidence angle (α_{95}).
(1) Reversal test (McFadden and McElhinny, 1990),: $\gamma_0 = 7.9 < \gamma_c = 8.3$, B class, pass at 95% confidence level.
(2) McElhiny's fold test (1964): $N = 8$, $k_s/k_g = 137.1/16.4 = 8.36 > F(2*(n\ 2-1),(n\ 1-1)) = F(14,14) = 6.13$, at the 99% confidence level fold test is positive.
(3) Fold test is positive at the 99% confidence level (McFadden, 1990), $\xi_2 = 6.136$ before and $\xi_2 = 0.160$ after tilt corrections, critical ξ, at 99% = 4.562 and 95% = 3.298.
(4) Watson and Enkin's fold test (1993): optimal concentration at 95.0 ± 4.1% unfolding percentage.
(5) Paleomagnetic pole: $\lambda = 18.9°$N, $\Phi = 208.4°$E, $A_{95} = 3.9$ (VGP calculated from the directions tilt corrected coordinates).
(6) Paleomagnetic latitude: 16.5 ± 3.9° S.

Table 4
The Late Triassic Paleomagnetic data of the Component C.

Site ID	n_1,n_2/N	In situ		Tilt corrected		α_{95}	k	Pole position	
		D_g (°)	I_g (°)	D_s (°)	I_s (°)			λp (°N)	ϕp (°E)
ZL/1,2	3,3/20	134.6	−45.8	69.4	−43.3	12.7	33.0	3.1	206.7
ZL/03	5,1/12	38.5	−8.7	42.3	−22.1	9.7	68.0	31.3	214.2
ZL/4,5	5,0/21	41.4	−17.5	47.6	−29.9	9.8	130.4	24.8	213.1
JR/1,2	1,5/18	36.2	9.4	46.4	−38.7	13.9	24.9	21.0	218.6
JR/3,4	3,3/16	31.7	1.3	48.0	−48.0	7.6	90.5	14.6	222.5
JR/6,7,8	7,1/31	58.6	−13.7	61.1	−17.5	10.8	33.6	18.9	198.0
All sites mean	6	49.4	−14.1	52.3	−33.6	33.4 (12.4)	5.0 (30.1)	19.6	211.8

Notes are as given in Table 3.
(1) McElhiny's fold test (1964): $N = 6$, $k_s/k_g = 30.1/5.0 = 6.02 > F(2*(n\ 2-1),(n\ 1-1)) = F(10,10) = 5.85$, at the 95% confidence level fold test is positive.
(2) Fold test is positive at the 95% confidence level (McFadden, 1990), $\xi_2 = 3.021$ before and $\xi_2 = 1.513$ after tilt corrections, critical ξ, at 95% = 2.862.
(3) Watson and Enkin's fold test (1993): optimal concentration at 107.0 ± 12.2% unfolding percentage.
(4) Paleomagnetic pole: $\lambda = 19.6°$N, $\Phi = 211.8°$E, $A_{95} = 10.7$ (VGP calculated from the directions tilt corrected coordinates).
(5) Paleomagnetic latitude: 18.4 ± 10.7°S.

McFadden fold test at the 99% confidence level ($\xi_2 = 6.130$ before, and $\xi_2 = 0.160$ after tilt corrections, critical value $\xi = 4.526$ at 99% and $\xi = 3.298$ at 95% confidence level) (McFadden, 1990), and the Watson and Enkin test gives an optimal concentration at 95.0 ± 4.1% unfolding percentage (Watson and Enkin, 1993) (Fig. 6). Its corresponding paleopole is at 18.9°N, 208.4°E ($A_{95} = 3.9°$).

The Component C was isolated in 6 sites (37 specimens) from the Late Triassic Gyangrang and Zhuglung Formations. The site-mean direction is $D_g = 49.4°$, $I_g = -14.1°$ ($k_g = 5.0$, $\alpha_{95g} = 33.4°$, $n = 6$ sites), and $D_s = 52.3°$, $I_s = -33.6°$ ($k_s = 30.1$, $\alpha_{95s} = 12.4°$, $n = 6$ sites) before and after tilt corrections. This site-mean direction passed the McElhinny's (1964) (($k_s/k_g = 30.1/5.0 = 6.02$) > (F (10, 10) = 5.85)) and McFadden's (1990) ($\xi_2 = 3.021$ before, and $\xi_2 = 1.513$ after tilt corrections, critical value $\xi = 2.862$) fold tests at the 95% confidence level (Fig. 6d, Table 2; please see Table 3 in the Supplementary material for detailed information). The Watson and Enkin test gives an optimal concentration at 107.0 ± 12.2% unfolding percentage (Watson and Enkin, 1993) (Fig. 6). The corresponding Late Triassic paleopole for the Lhasa Terrane is at 19.6°N, 211.8°E ($A_{95} = 10.7°$) (see Table 4).

6. Discussion

6.1. Age of magnetization Component C

As described in previous sections, we conducted a paleomagnetic study into the newly discovered Triassic marine carbonate formations of the Lhasa Terrane. The geologic age of the rocks was well-determined by the biostratigraphic studies of conodonts. After the removal of two post-folding age components (Component A and Component B), the characteristic remanences component (Component C) was isolated from 8 sites (47 specimens) of the Early-Middle Triassic Formation, and from 6 sites (37 specimens) of the Late Triassic Formations.

The Component C of the Early-Middle Triassic rocks passed both reversal (95% confidence level) and fold tests (99% confidence level), that of the Late Triassic rocks pass a fold tests (95% confidence level). Both rock magnetic experiments and thermal demagnetization indicated that the component C was carried by magnetite, which is capable of preserving an early remanent magnetization over a long period of time. Moreover, the new results are in accordance with the published late Paleozoic paleomagnetic data and bears no resemblance to the paleopoles of younger age for the Lhasa Terrane. This Triassic paleomagnetic results met all the reliability criteria reported by Van der Voo (1990). Thus, we propose that the component C should be interpreted as a primary remanent magnetization of the Triassic Period and calculating the corresponding paleomagnetic poles for the Lhasa Terrane.

6.2. Paleogeographic implications of the paleomagnetic results

We conducted a series of paleomagnetic investigations on the Late Paleozoic and Early Mesozoic rocks from the central Lhasa Terrane. Similarities in the characteristics of the demagnetization and negative field tests suggest that a widespread remagnetization most likely occurred in the Late Paleozoic rocks in central Lhasa Terrane (Guo, 2009; Cheng, 2012). Ran et al. (2012a) published a five sites mean direction of the Early Permian for the Lhasa Terrane, which included three sites of the preliminary results listed in the MS thesis (Guo, 2009). This combined direction, corresponding to the pole at 24.5°N, 219.3°E ($A_{95} = 7.8°$), which are in accordance with the Early-Middle and Late Triassic paleomagnetic poles present in this paper for the Lhasa Terrane. We suggest that the Lhasa Terrane was paleogeographically located at mid-to-low latitudes of the southern hemisphere during the Early Permian to Late Triassic period. The Lhasa Terrane might have become amalgamated with the main body of Eurasia in the Cretaceous Period (Chen et al., 1993; Aitchison and Davis, 2001; Yang et al., press).

Cheng et al. (2012, 2013) have reported paleomagnetic data from the Late Carboniferous, Early-Middle Permian, Late Permian and Early Triassic strata from the central part of the Qiangtang Terrane. The primary origin of the magnetization is confirmed by positive fold-, reversal-, and baked contact tests. These results indicate that the Qiangtang Terrane was paleogeographically located at mid-to-low latitudes of the southern hemisphere during the Late Carboniferous-Early Triassic period. As shown in Fig. 7 and Table 5, the paleomagnetic pole positions of the Qiangtang and Lhasa Terranes in the Early Triassic were located adjacent to each other. The differences in their relative rotation and latitude are close to zero in the confidence range of the data, indicating that the Qiangtang and Lhasa terranes were likely adjacent to each other during the Early Triassic.

Ran et al. (2012b) report a new Triassic paleomagnetic data with positive fold test for the Tethyan Himalaya (the corresponding paleopole at 34.5°N, 282.2°E with $A_{95} = 11.7$), suggesting the close affinity between the Tethyan Himalaya and cratonic India since the Triassic at paleolatitude of 25.0°S (reference location: 28.7°N/86.2°E). This paleopole is statistically distinguishable from the Triassic paleopoles of the Lhasa and Qiangtang terranes. So we propose that the Qiangtang and Lhasa terranes might not have been adjacent to the northern margin of Gondwanaland and situated in an isolated position between Eurasia and Gondwanaland since the Early Triassic Period.

6.3. Constraints on the evolution of the Meso-Tethys

Song et al. (2012) obtained two Late Triassic paleopoles from sandstone specimens of the Tumengela Formation and Zala Formation in the Qiangtang block: one at 59.2°N, 179.2°E, with $A_{95} = 4.1$ and the other at 65.1°N, 184.4°E, with $A_{95} = 3.7°$. These paleopoles correspond to paleolatitudes of 30.3 ± 4.1°N and 29.7 ± 3.7° N (reference point: 34.1°N, 92.4°E). In contrast, Song et al. (2015) obtained one Late Triassic paleopole from the Upper Triassic Jiapila Formation volcanic rocks on the northern edge of the Qiangtang block of Central Tibet: 64.0°N, 174.7°E, with $A_{95} = 6.6$ ($N = 29$), corresponding to paleolatitude of 33.5 ± 6.6°N (reference point: 34.1°N, 92.4°E). Since both volcanic result (Song et al., 2015) and the sedimentary result (Song et al., 2012) fulfill all seven criteria and are statistically indistinguishable. So we used the volcanic result (Song et al., 2015) for further discussion.

Taking the paleomagnetic pole of the Lhasa Terrane as a reference pole, we calculated the corresponding expected paleomagnetic parameters for the reference point of the Qiangtang Terrane (located at 34.1°N, 92.4°E). This comparison indicates that the Qiangtang Terrane moved rapidly northward relative to the Lhasa Terrane during the Early to Late Triassic Period (about 25.4 ± 9.9°), accompanied by a counter-clockwise rotation of approximately 44.8 ± 8.7°. According to the paleomagnetic results, we suggest that the Meso-Tethys oceanic basin, potentially represented by the BNS between the Lhasa and Qiangtang terranes, opened up in the Early Triassic and expanded rapidly during the Early to Late Triassic Period. This hypothesis agrees with the Tarenben oceanic-island basalts, characterized by pelagic radiolarian silicalite facies and the submarine eruptive facies, and SSZ-type ophiolite along the Bangong-Nujiang suture zone of the Middle Triassic – Middle Jurassic ages (Wang et al., 2005; Pan et al., 2006; Shi, 2007; Li et al., 2011). The width of the Meso-Tethys ocean basin reached at least 2500 km in the Late Triassic Period. The rapid expansion of the Ocean drove movement of the Qiangtang Terrane northwards which promoted its subsequent amalgamation with the southern margin of Eurasia.

7. Conclusions

This paper presents the new Triassic paleomagnetic poles form the central Tibet (the Early-Middle and Late Triassic of Lhasa Terrane). Our new results, together with the previous published poles, indicate that:

(1) The Lhasa Terrane maintained a relative stable latitude in the southern hemisphere (16.5 ± 3.9°S and 18.4 ± 10.7°S) during the Triassic Period before moving northward and amalgamating with the main body of Eurasia.

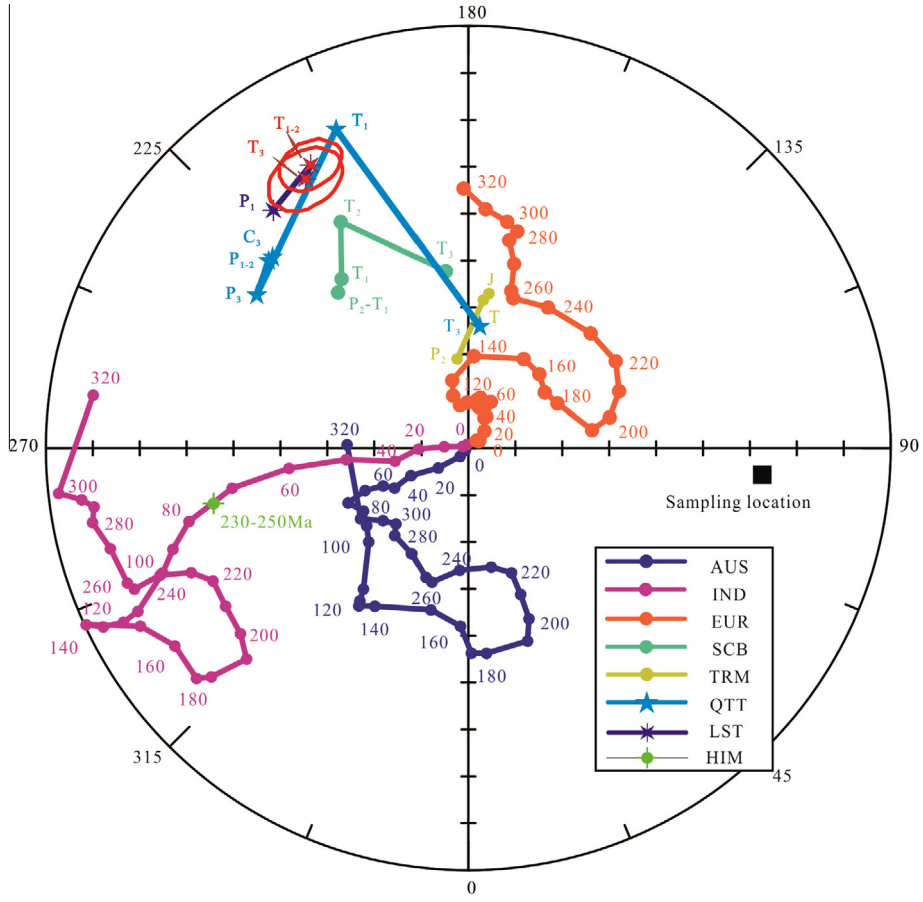


Fig. 7. Stereographic projection of the paleomagnetic poles discussed in this paper. Apparent polar wander path (APWP) of Eurasian (EUR), Australian (AUS), and India (IND) (Torsvik et al., 2008); APWP of Tarim Block (TRM), South China Block (SCB) (Huang et al., 2008); poles of Qiangtang Terrane (QTT) (Cheng et al., 2012, 2013; Cheng, 2012; Song et al., 2015); pole of Himalaya Terrane (HIM) (Ran et al., 2012b); Early Permain pole of the Lhasa Terrane (LST) (Ran et al., 2012a); red symbols (with A_{95} confidence circles) showing the Early-Middle Triassic and Late Triassic poles of the Lhasa Terrane from this study. See Table 5 for the values and reference. Black squares: sampling location.

Table 5
Paleopoles from Lhasa Terrane and adjacent terranes/continents.

Region	Age	Lat./Lon.	*n*/*N*	Plat./Plon.	A_{95}	P_λ	Test	Reference
Qiangtang	T_3	34.1/92.4	29/254	64.0/174.7	6.6	27.5 N	F	Song et al. (2015)
Qiangtang	T_1	33.7/86.7	2/15	16.5/202.5	6.7	13.8 S	–	Cheng et al. (2012)
Qiangtang	P_3	33.7/86.7	5/30	34.4/234.1	9.3	18.6 S	F,R,BC	Cheng et al. (2012)
Qiangtang	P_{1-2}	33.7/86.7	5/46	31.7/226.8	12.5	17.8 S	F,R,BC	Cheng et al. (2012)
Qiangtang	C_3	33.7/86.7	2/18	31.8/225.7	2.9	17.2 S	–	Cheng et al. (2012)
Lhasa	T_3	30.86/84.71	6/37	19.6/211.8	10.7	18.4 S	F	This study
Lhasa	T_{1-2}	30.86/84.71	8/47	18.9/208.4	3.9	16.5 S	F,R	This study
Lhasa	P_1	30.9/88.3	5/28	24.5/219.3	7.8	19.6 S	F	Ran et al. (2012a)
Himalaya	T_1	28.7/86.2	10/52	34.5/282.2	11.7	22.6 S	F	Ran et al. (2012b)
Southern China	T_3	–	2/–	52.0/187.3	–	16.8 N	–	Huang et al. (2008)
Southern China	T_2	–	4/–	34.7/209.4	9.1	6.3 S	–	Huang et al. (2008)
Southern China	T_1	–	9/–	45.0/216.8	3.9	2.5 S	–	Huang et al. (2008)
Southern China	P_2–T_1	–	7/–	46.8/219.9	6.4	2.4 S	–	Huang et al. (2008)
Tarim	J	–	2/–	56.8/172.5	–	26.6 N	–	Huang et al. (2008)
Tarim	T	–	3/–	58.3/174.3	15.7	26.1 N	–	Huang et al. (2008)
Tarim	P_2	–	5/–	70.9/187.3	10.3	25.1 N	–	Huang et al. (2008)

Note: Lat.: latitude, Lon.: longitude; Plat.: pole latitude; Plon.: pole longitude; *n*/*N*: number of sites/samples accepted; A_{95}: the radius that the mean pole lies within 95% confidence; P_λ: paleolatitude for a reference point at 30.86°N/84.71°E; F: fold test; R: reversal test; BC: baked contact test.

(2) Both the Qiangtang and Lhasa terranes might not have been adjacent to the northern margin of Gondwanaland and have been situated in isolation between Eurasia and Gondwanaland since the Early Triassic Period.
(3) The Meso-Tethys, potentially represented by the BNS, opened up in the Early-Middle Triassic and expanded during the whole Triassic period.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jseaes.2016.02.006.

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